

Diesel engine

12V4000G21

Spare parts catalog for product no.

526101438

Printed in Germany

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This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

Subject to alterations and amendments.

Preface

This catalog is published for parts requisitioning purposes and provides assistance in engine disassembly and reassembly. It contains a complete list of all engine components, including those which are not available as replacement parts.

The parts originate from the Engine Design File and document the product's construction status.

Catalog Arrangement

The catalog is subdivided into functional groups and functional group segments (FG/FGS).

The headings of the illustration pages contain the following information:

- ◆ TYPE
- ◆ SERIAL NO.
- ◆ FG/FGS DESTINATION
- ◆ FG/FGS NO.
- ◆ BOM NO.

Explanation of Columns

- ◆ POS Provides a cross-reference from illustration to text section. Non-illustrated parts are indicated by the suffix "Z" after the C/N.
- ◆ PART NO. This is the valid number for ordering the associated part.
- ◆ DESIGNATION The precise part name.
- ◆ QTY (Quantity of Unit) This column indicates the quantity of this part required for the functional group.
- ◆ MEASUREMENT
- ◆ FN (Footnote) Provides additional parts information. For clarification of footnote symbols, see Page 3/2.
- ◆ REMARKS This column is intended for supplementary information.

Amendments Service

Part No. changes and amendments which occur after publication of the catalog will not be published. We therefore recommend updating of the catalog part numbers by the end user. This information is provided, for instance, in order confirmations and/or on equipment labels.

Footnote Symbols Identification

AB Supplied as a complete assembly. Repair by manufacturer only.

AC Not scheduled as a spare part. Availability must be checked.

EM Emission-relevant part. In the case of all marine engines with IMO certification it should be noted that the identification numbers of the emission-relevant parts on the engine must match the details given in the technical file. When ordering, quote EM code with order no. E.g: 535 030 04 17 EM.

KD Group balancing required.

KE Individual parts not obtainable through the parts supply system.

KF Calibrated part.

M Parts having different repair sizes. Dimensions are listed in the Tolerances and Wear Limits List.

P Bought-out components.

SB Group of parts which are welded or brazed together.

UA Indicate dimensions when ordering.

UB Only supplied complete or as a complete set.

Ordering Procedure

The following information is essential for spare-part order processing:

- ◆ TYPE
- ◆ SERIAL NO.
- ◆ FG/FGS

- ◆ POS
- ◆ PART NO.
- ◆ DESIGNATION
- ◆ QUANTITY
- ◆ PAGE

When ordering replacement parts for ancillary equipment such as injection pumps, fuel pumps, etc., give both model and part number of the part.

Phone orders should be confirmed in writing to prevent errors.

Ordering Example

Type	Serial number	FG/FGS	BOM	POS.	Part number	Designation	Qty.
8V4000G60	524100298	075 05	070.00075	1	0010104351	Injector	8
		083 05	524.903-30	10	5240900252	Fuel Filter	2

Factory Repairs

Parts shipped to MTU for repair should have an engine serial-number identification (tag or label, for example).

Exchange Components (Reman)

Exchange components are parts overhauled by the manufacturer, providing the same properties with regard to service life and quality as new parts.

Components which are included in the Reman exchange procedure, are marked by the following letters added before the item number:

E 524 010 35 20 = New component in exchange procedure.

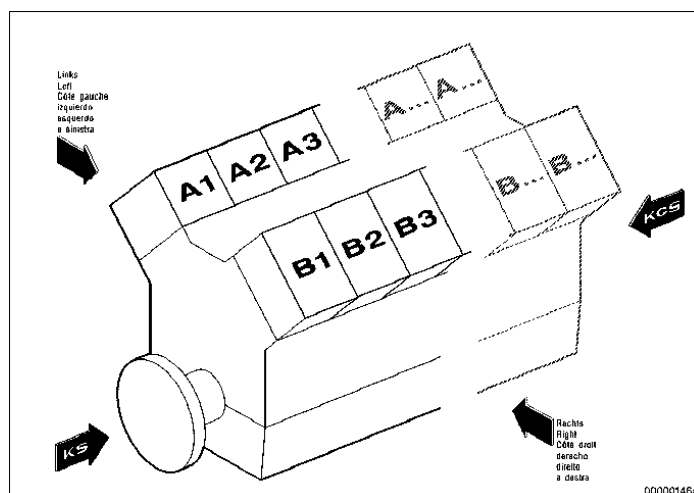
R 524 010 35 20 = Reman component in exchange procedure.

Exchange Procedure

With the introduction of exchange components, the so-called core deposit will also be charged for new parts. The core deposit will be indicated separately on the invoice and the respective value will be credited upon return of reusable components.

For further information refer to Service Information 01/S0028.

Engine Side And Cylinder Designations



Engine sides are always designated as viewed from driving end.

The left bank of cylinders is marked 'A' and the right bank 'B' (according to DIN ISO 1204).

The cylinders of each bank are numbered consecutively, starting with No. 1 at the drive end.

Numbering of other engine components is also from the drive end, starting with No. 1.

Designations and Abbreviations

Driving end = KS

Free end = KGS

Left side

Right side

Upper side

Lower side

Turbocharger = ATL

Overview functional groups

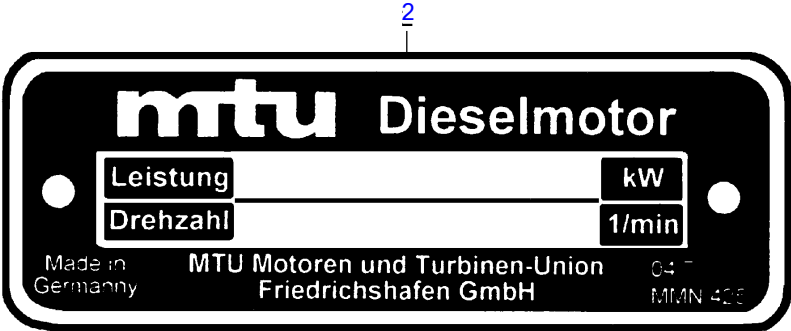
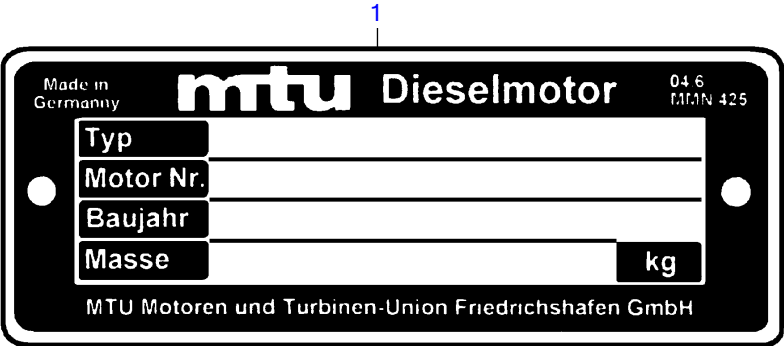
FG	FG DESIGNATION
004	LABELS AND MISCELLANEOUS MATERIALS
011	CRANKCASE
013	CYLINDER LINER
014	OIL PAN
015	ENGINE MOUNTING
016	END HOUSING, MAIN PTO END AND AUX PTO END
018	CRANKCASE BREATHER
024	GEARCASE, AUX PTO END
031	CRANKSHAFT ...
032	CRANKSHAFT SUPPORT BEARINGS, MAIN PTO END
035	CRANKSHAFT, PTO, AUX PTO END
036	CRANKSHAFT TRANSPORTATION LOCKING DEVICE
037	PISTON
038	CONROD
041	CYLINDER HEAD
042	CYLINDER HEAD ATTACHMENTS
051	CAMSHAFT
052	CAMSHAFT DRIVE
053	CAMSHAFT AXIAL ALIGNMENT BEARING
054	ROLLER TAPPETS, PUSH RODS
055	VALVE DRIVE
073	INJECTION PUMP / HP PUMP
075	FUEL INJECTOR
077	HP FUEL LINE / HP LINE
081	FUEL PIPEWORK BEFORE DELIVERY PUMP
082	PIPEWORK BETWEEN DELIVERY PUMP AND FILTER
083	FUEL FILTER
086	LEAK-OFF FUEL LINE
101	TURBOCHARGER (H.P)
111	CHARGE AIR COOLING, IF REQUIRED, LEFT SIDE
121	AIR FILTER
124	AIR SUPPLY TO CYLINDERS
125	AIR SYSTEM, TURBOCHARGER - INTERCOOLER
141	EXHAUST SYSTEM, CYLINDER HEAD - EXH. MANIFOLD
172	ELECTRIC STARTING
181	LUBE OIL PUMP AND DRIVE
183	OIL FILTRATION/COOLING
184	CRANKCASE-INTERNAL OIL SUPPLY
185	TURBOCHARGER OIL SUPPLY
202	HIGH-TEMPERATURE CIRCUIT
203	LOW-TEMPERATURE CIRCUIT
206	COOLANT PIPEWORK WITH THERMOSTAT
213	GENERATOR
231	ENGINE MOUNTING
507	GENERAL WIRING, ENGINE/GEARBOX/PLANT
511	ALARM SYSTEM

BOM NO. _____

004 05
INSTALLATION OF LABELS
000.00115

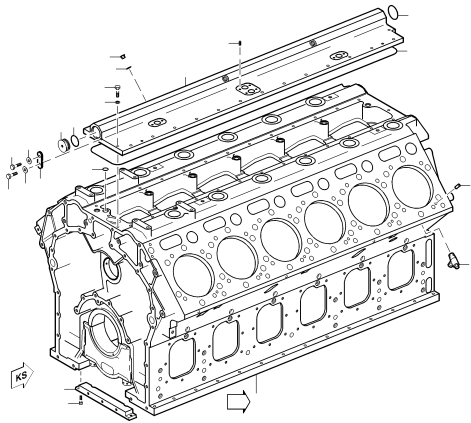


00400006a



00400006a

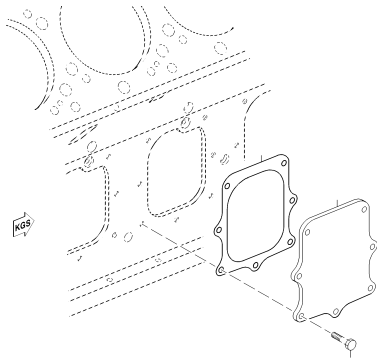
POS.	PART NO.	DESIGNATION	QTY.	MEASUREMENT
1	700425000135	PLATE	1	04.6
2	700425000136	PLATE	1	04.7



01100235a

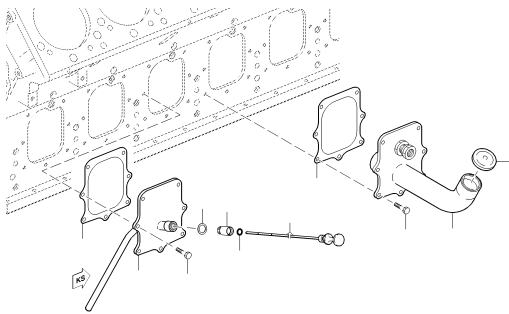
011
CRANKCASE
 010.00382

05



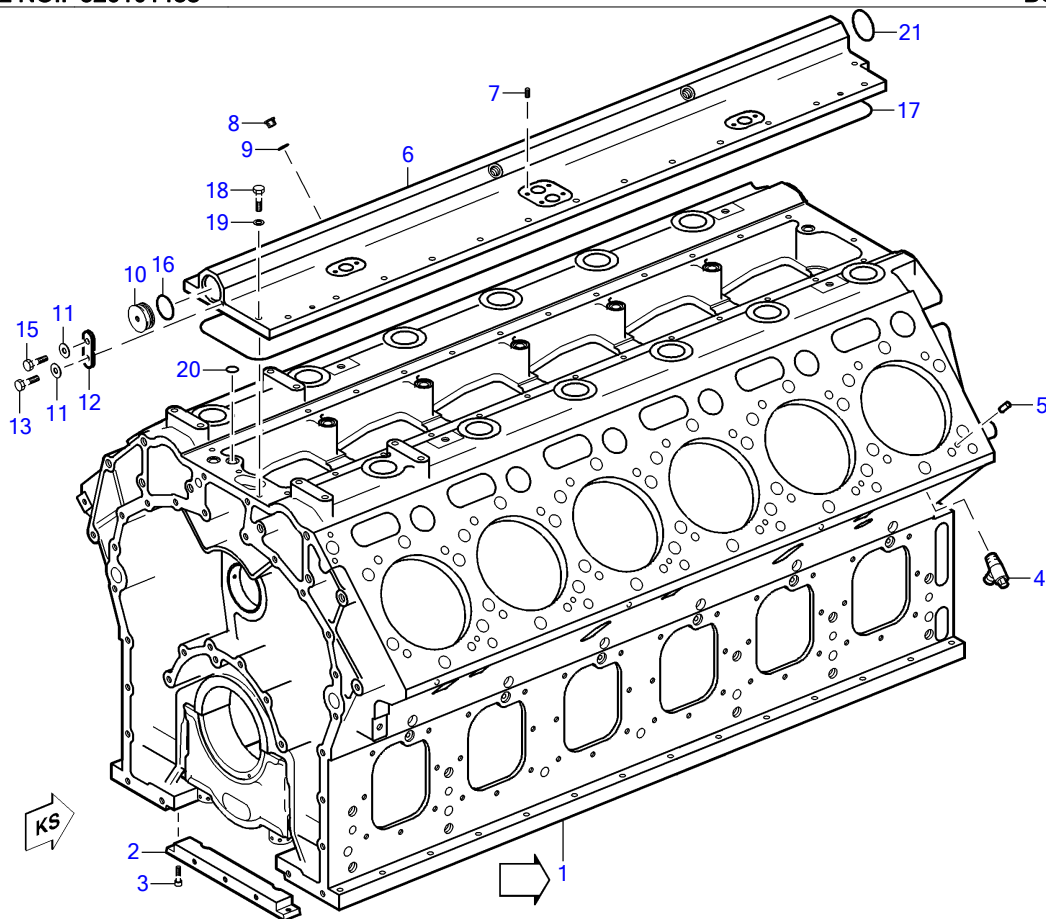
E0113048b1

011 30
**INSPECTION PORT COVER AND ANTI-EXPLOSION VALVE, IF
 REQUIRED, LEFT SIDE**
 010.00382



01100179a

011 40
FLANGE AND GUIDE TUBE FOR OIL DIPSTICK
 010.00390



01100235a

POS.	PART NO.	DESIGNATION	QTY.	MEASUREMENT
1	5260101305	CRANKCASE	1	
2	5240110440	STRIP	1	
3	000912008115	SCREW	2	M 8 X 16
4	0049976736	DRAIN VALVE	2	
5	000007010124	DOWEL PIN	12	10 M6 X 20
6	5260101114	COVER	1	
7	000835010056	STUD	8	M 10 X 25
8	000908014001	PLUG	1	M 14 X 1,5
9	007603014102	SEALING RING	1	A 14 X 18
10	735084050000	PLUG	1	50
11	007349010002	WASHER	2	10,5
12	5249950059	STRAP	1	
13	000933010089	SCREW	1	M 10 X 20
15	000933010198	SCREW	1	M 10 X 16
16	700429042000	O-RING	1	42 X 5
17	5269970345	O-RING	1	1057X5
18	000933010093	SCREW	28	M 10 X 35
19	000125010524	WASHER	28	A 10,5
20	700429018005	O-RING	8	18 X 4
21	700429055001	O-RING	1	55 X 5